№01 · TCEX TECHNOLOGIES, CHENNAI

TraceCore.

Lab records that stay trustworthy in the quantum era.

Stakeholder meeting, 15 October 2026 · [Presenter name, role]

№02 · THE STAKE

Every approved medicine and medical device rests on a lab record.



If that record is altered, or leaked, the whole chain breaks.

№03 · WHAT A LIMS IS

A LIMS is the lab's system of record.

Sample

09:12 analyst A registered

Test

10:42 analyst A result entered

Review

14:05 QA lead reviewed

Report

16:30 manager released

Archive

16:31 system locked

EVERY STEP LOGGED

Who did it, what changed, when, and why. Used by pharma, medical device, CRO and food testing labs.

THE QUESTION EVERY AUDITOR ASKS

“Can you prove this record was never altered?”

№04 · THE PROBLEM TODAY

Paper and spreadsheets cannot prove a record is untouched.

TODAY, IN MOST LABS

01 Records sit across paper registers, Excel files and email

02 Values are re-typed by hand from instruments

03 Edits leave no trace of who changed what

04 Audits take days of reconstruction

2026 · US FDA WARNING LETTERS

22

Indian sites received warning letters, at least.

About half cited data integrity or incomplete lab records.

№05 · HARVEST NOW, DECRYPT LATER

Data stolen today can be read tomorrow.

01 · TODAY

Attackers copy encrypted data and store it.

02 · YEARS PASS

The copy sits untouched, waiting.

03 · QUANTUM ARRIVES

Public-key encryption (RSA, ECC) breaks. The data opens.

MOSCA'S RULE

secrecy period + migration time > time until quantum → you are already late.

№06 · WHY LAB DATA IS A PRIME TARGET

Lab data must stay secret and trusted for decades.

01	**Study protocols and research data**	Intellectual property for the life of the product
02	**Sponsor and client information**	Commercial contracts and unreleased products
03	**Regulated reports**	Kept for years under GLP and GMP retention rules
04	**Signing and identity keys**	They vouch for every record in the archive

№07 · QUANTUM RISK IS ALSO A FORGERY RISK

A broken signature means a record can no longer prove anything.

TODAY

A digital signature proves who approved a record.

RSA, ECDSA

WITH A CAPABLE QUANTUM COMPUTER

The same signature can be forged.

An archived approval would then prove nothing.

For regulated labs, trust in the record matters as much as secrecy.

№08 · THE MIGRATION CLOCK HAS STARTED

Standards and national deadlines are already set.

2024	NIST	Post-quantum standards published: FIPS 203 (ML-KEM), 204 (ML-DSA), 205 (SLH-DSA)
2027	INDIA	PQC rollout begins in critical information infrastructure
2028	INDIA	Enterprises begin structured preparation
2030	NIST	RSA-2048 and similar algorithms deprecated
2033	INDIA	Full national adoption targeted
2035	NIST	Quantum-vulnerable algorithms disallowed

INDIA'S TASK FORCE CALLS FOR

Crypto inventories, vendor crypto bills of materials, and quantum-safe procurement.

SOURCE — NIST IR 8547 (DRAFT); INDIA NATIONAL QUANTUM MISSION TASK FORCE REPORT, FEB 2026

№09 · OUR APPROACH

TraceCore builds post-quantum protection into the lab record.

LAYER 01

Access

Passwordless sign-in for each user

STATUS [CONFIRM]

LAYER 02 · ML-KEM

Transport

Data moving between lab, instruments and server

STATUS [CONFIRM]

LAYER 03

Storage and keys

Stored records and encryption keys

STATUS [CONFIRM]

LAYER 04 · ML-DSA

Signatures and audit trail

Approvals and every logged change

STATUS [CONFIRM]

Crypto-agile: algorithms swap without a rebuild [CONFIRM]

Runs on the lab's own servers: the lab keeps its data and keys [CONFIRM]

№10 · WHAT THE PLATFORM DOES

One connected record from sample to archive.

01

Samples

Registration, unique ID, storage location, status

02

Studies

Protocol, study director, sponsor, lifecycle

03

Testing

Instrument data captured directly, limits checked `[CONFIRM which instruments]`

04

Quality

Deviations, CAPA, approvals

05

Reports

Version control, controlled release, retrieval

06

Integrity

Role-based access, server-set timestamps, a reason for every change, original value always kept

№11 · LIVE DEMO

One sample, end to end.

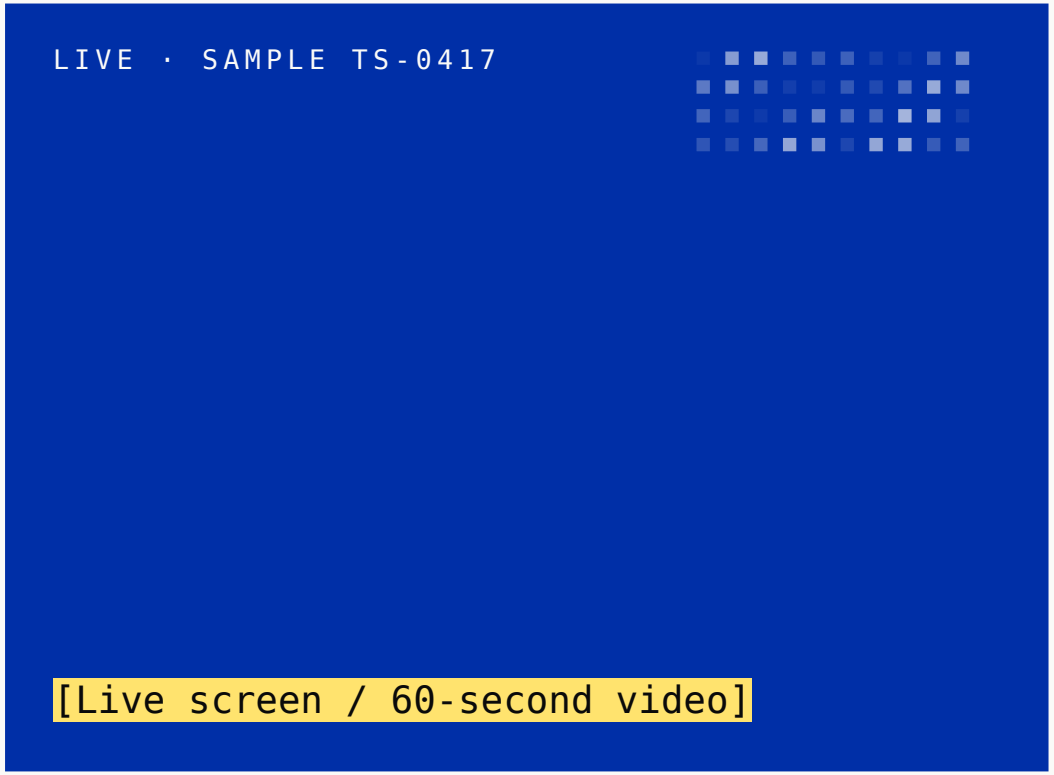
01 Register a sample. The server sets the date; it cannot be changed.

02 Capture a result from an instrument.

03 Try to edit it. The system demands a reason.

04 Open the audit trail: old value, new value, who, when, why.

05 Approve with an electronic signature [show the PQC signature if built]



№12 · HOW A LAB SHOULD JUDGE ANY LIMS

Five questions every lab should ask its software vendor.

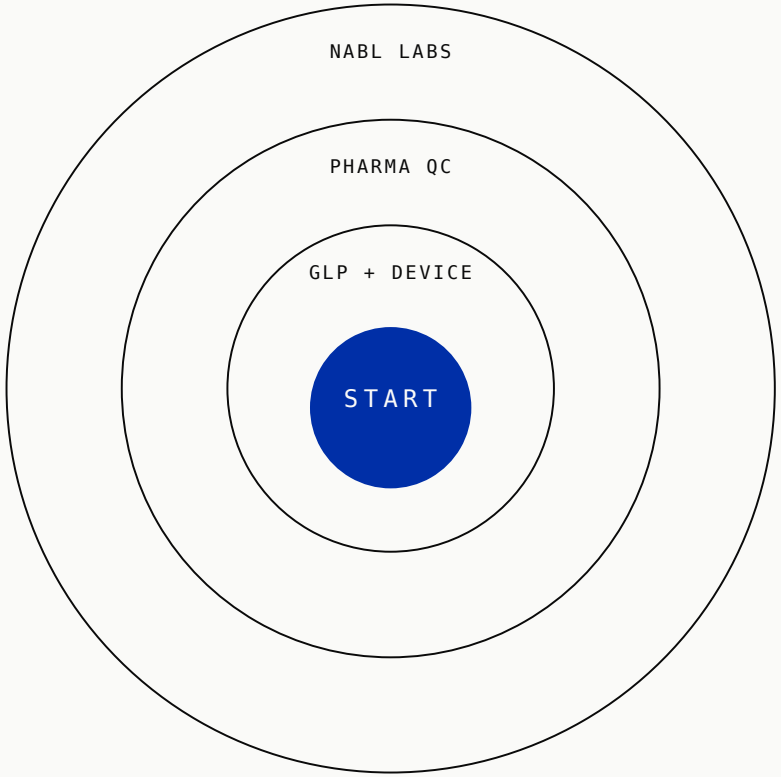
QUESTION	TRACECORE
Can every record be traced from sample to archive?	**Yes, linked end to end**
Can I see who changed what, why, and who approved it?	**Yes, full audit trail**
Are roles and duties separated?	**Yes, role-based access**
Where does my data live, and who holds the keys?	**On the lab's servers `[CONFIRM]`**
How will the system move to post-quantum cryptography?	**Built in `[CONFIRM status]`**

№13 · WHERE WE START

We start where data must stay protected the longest.

61	certified GLP pre-clinical facilities
~89	medical device testing labs (CDSCO)
~10,500	pharma units, each with a QC lab
6,815	NABL-accredited testing labs, next

First deployment: **[MDR, if naming is cleared]**



SOURCE — DST GLP LIST, CDSCO REGISTER, NABL PORTAL, READ 1 OCT 2026; PHARMA UNITS: INDUSTRY ESTIMATE

№14 · WHY THIS BELONGS IN QUANTUM VALLEY

Quantum Valley builds the compute. We protect the data it could expose.

COMPUTE

Amaravati will host an IBM Quantum System Two

Pharma is among its target applications.

156

QUBITS · HERON PROCESSOR

EXPOSURE

The same sector holds the data quantum could expose

Pharma and device labs keep records for decades.

15 yrs

LIFE OF A PRODUCT'S DATA

PROCUREMENT

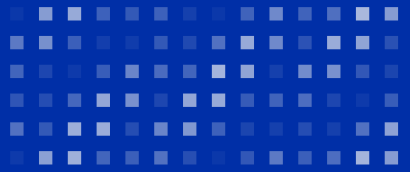
India will require quantum-safe procurement

TraceCore can be an early reference application.

2027

PQC ROLLOUT BEGINS

Built in South India, for Indian regulated labs.



Regulated science runs on trust.
We are making that trust last.

№15 · THE ASK

What we are asking for.

01	Access to Quantum Valley testbeds to benchmark our post-quantum layer
02	Expert review of our cryptographic design by ecosystem partners
03	Introductions to 2 or 3 pilot labs in Andhra Pradesh
04	[CONFIRM: incubation, grant or funding ask, with amount]

№16 · APPENDIX A1 · REGULATORY CONTEXT

The rules lab records live under.

OECD GLP	Pre-clinical study conduct and records
21 CFR Part 11	Electronic records and signatures for US FDA submissions
ALCOA+	Attributable, legible, contemporaneous, original, accurate, plus complete, consistent, enduring, available
Revised Schedule M	Indian GMP: a record of every data change, in force for all units since 1 Jan 2026

No body certifies LIMS software. Labs validate the configured system; vendors supply validation documents.

№17 · APPENDIX A2 · STATUS OF EVERY CLAIM

Where each capability stands today.

	BUILT	TESTED	IN PROGRESS	PLANNED	EXT. REVIEWED
Audit trail and controlled correction	☐	☐	☐	☐	☐
Electronic signatures	☐	☐	☐	☐	☐
ML-KEM key establishment	☐	☐	☐	☐	☐
ML-DSA signatures	☐	☐	☐	☐	☐
Passwordless sign-in (KratosId)	☐	☐	☐	☐	☐
Instrument integration	☐	☐	☐	☐	☐
AI compliance assistant	☐	☐	☐	☐	☐